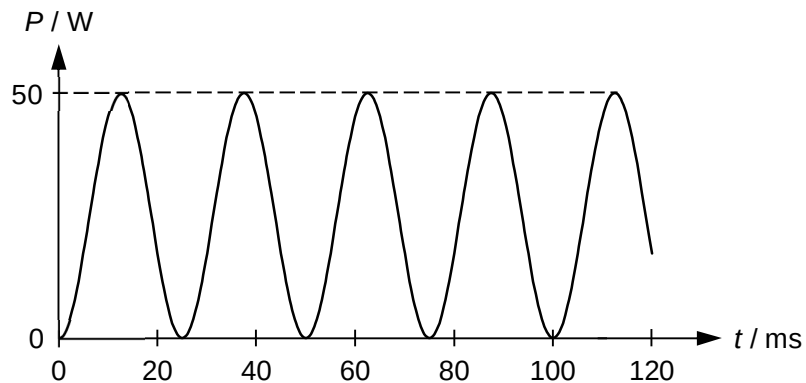


- 28** A sinusoidal alternating supply is connected to a  $10\ \Omega$  resistor in series. The graph shows how the power  $P$  dissipated across the resistor varies with time  $t$ .



What are the frequency  $f$  and the root-mean-square current  $I_{\text{r.m.s.}}$  of the alternating supply?

|          | $f / \text{Hz}$ | $I_{\text{r.m.s.}} / \text{A}$ |
|----------|-----------------|--------------------------------|
| <b>A</b> | 20              | 1.6                            |
| <b>B</b> | 40              | 1.6                            |
| <b>C</b> | 20              | 2.2                            |
| <b>D</b> | 40              | 2.2                            |

20 The energy loss in a thin metal film is of the order of magnitude  $10^{-10}$  eV